TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 3

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- Read each question **carefully**
- Answer **all** questions
- Write your answers **clearly** in the spaces provided
- **Check** your answers.

Information:

- This examination has **two** sections. These are clearly labelled
- You are **not allowed** to use a calculator for Section A
- You **are allowed** to use a basic calculator for Section B
- The **maximum** mark for this examination is 60
- The marks available for each question are shown in **bold** beneath each question.

Exam paper

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

- Assessment: Past Paper 3 (Non-Calculator & Calculator)
- Complete stepped calculations, methods & mark scheme codes [MI/AI]
- Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)
- Pass standard: 34 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Section A	15		
Section B	45		
Total marks	60		

MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and NHS healthcare staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

© Training Qualifications UK 2023 All Rights Reserved

Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 0.238 + 0.05

Calculate: 0.72 × 0.4 = 0.288

Answer 0.288 ✓ [1 mark]

[1]

2. Work out 0.8 × 0.222

Calculate: 1.48 × 0.12 = 0.1776

Answer 0.1776 ✓ [1 mark]

[1]

Past paper

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

3. A company has 200 employees and 2 offices.

Employees are split across both offices as shown in the table below:

	Office A	Office B
Part-time employees	64	48
Full-time employees	16	72

One person is chosen at random.

What is the probability that the person chosen works full time in office B?

Give your answer as a percentage.

Step 1: Probability = 72 / 200

Percentage: (72 ÷ 200) × 100 = 36%

Answer 36% ✓ *[2 marks]* %

[2]

4. Round 39.4528 to the nearest whole number.

Use your rounded number to **estimate** the answer to 39.4528 × 2

Show your working.

Round 39.453 to 2 significant figures = 39

Double the rounded number: 39 × 2 = 78

Answer 78 ✓ *[2 marks]*

[2]

Past paper

 © Training Qualifications UK 2023 All Rights Reserved

5. Idris sees this mobile phone in a sale:

Mobile phone
Was £95.00
Now 12% off

How much money will Idris save buying the phone in the sale?

Step 1: Discount = 12% of £95

Calculation: 95 × 0.12 = £11.40

Answer £ **£11.40** [2 marks]

[2]

6. Work out $10^2 – 25 \times 4$

BIDMAS: Powers and Multiplication before Subtraction

$10^2 = 100$ and $25 \times 4 = 100$ --> $100 - 100 = 0$

Answer **0** [2 marks]

[2]

Past paper

Questions continue on the following page

 © Training Qualifications UK 2023 All Rights Reserved

7. Work out:

$$5\frac{1}{3}+4\frac{1}{5}$$

Give your answer as a mixed number.

Common denominator of 3 and 5 is 15:

5 1/3 = 5 5/15 and 4 1/5 = 4 3/15

Total = (5 + 4) + (5/15 + 3/15) = 9 8/15

Answer 9 8/15 ✓ [2 marks]

[2]

8. Alex has just been on a cruise:

- the cruise ship travelled 4320 miles
- it took 10 days of non-stop sailing.

Alex thinks the average speed of the cruise ship was 20 miles per hour.

Is Alex correct?

Show how you decide.

Total hours sailing = 10 days × 24 hours/day = 240 hours

Average speed = 4320 miles ÷ 240 hours = 18 mph

Comparison: 18 mph < 20 mph. Therefore No.

Answer No (18 mph < 20 ✓) [3 marks]

[3]

End of Section A.

Past paper

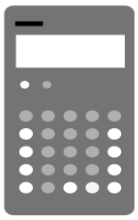
© Training Qualifications UK 2023 All Rights Reserved

Section B begins on Page 6.

Past paper

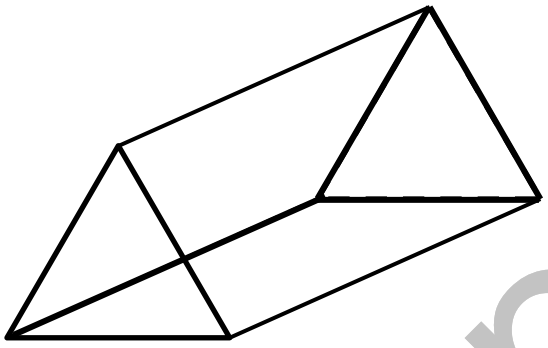
© Training Qualifications UK 2023 All Rights Reserved

Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. What is this shape?



Answer	Triangular prism ✓ [1 mark]

[1]

2. Put these numbers in order starting with the lowest:

1 025 600 –102 560 –85 956 1 102 560

Answer	-102 560 , -85 956 , 1 025 600 , 1 102 560 ✓ [1 mark]		
	Lowest	→	**Highest**

[1]

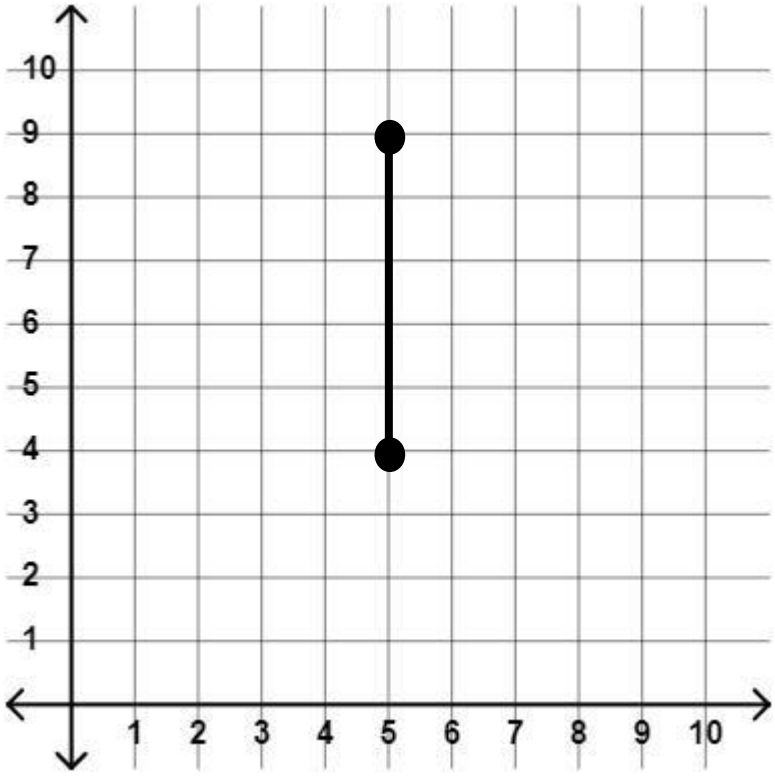
Past paper

© Training Qualifications UK 2023 All Rights Reserved

3. Ezra is going to put a shed in a garden.

The shed is in the shape of a square.

The coordinate grid shows where two corners and one edge of the square will go.



What are the possible coordinates for the other **two** corners of the square?

Shed width = 2 m (4 squares), length = 3 m (6 squares)

Opposite vertices from (6, 4) and (6, 9) --> (10, 4) and (10, 9)

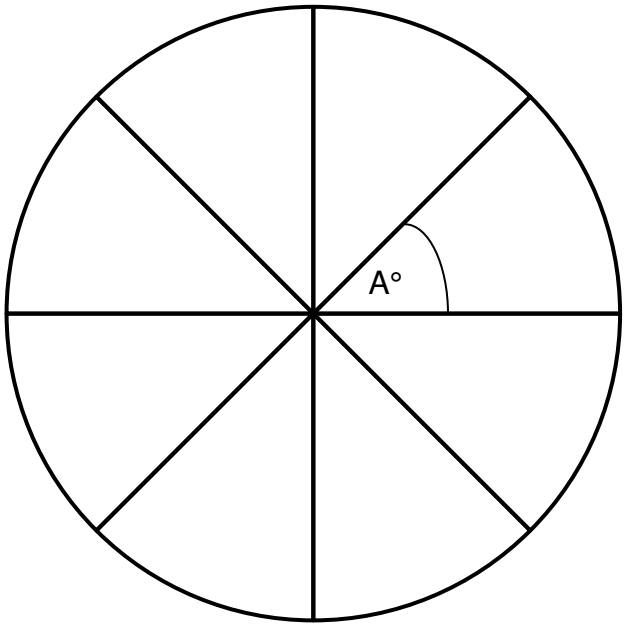
Answer (10, 4) and (10, 9) [2 marks]

[2]

Questions continue on the following page

 © Training Qualifications UK 2023 All Rights Reserved

4. A circle is divided into 8 equal sections as show below:



Calculate the value of A.

Full circle = 360°. Divided into 8 equal sectors:

Angle of each sector = 360° ÷ 8 = 45°

Answer 45° ✓ *[2 marks]* °

[2]

Past pap

© Training Qualifications UK 2023 All Rights Reserved

5. Write this fraction:

$$\frac{29}{40}$$

(a) as a decimal

(b) as a percentage.

Fraction 29/40: 29 ÷ 40 = 0.725

Percentage: 0.725 × 100 = 72.5%

Answer	**(a) Decimal:** 0.725	**(b) Percentage:** 72.5% ✓ [2 marks]

[2]

Past paper

Questions continue on the following page

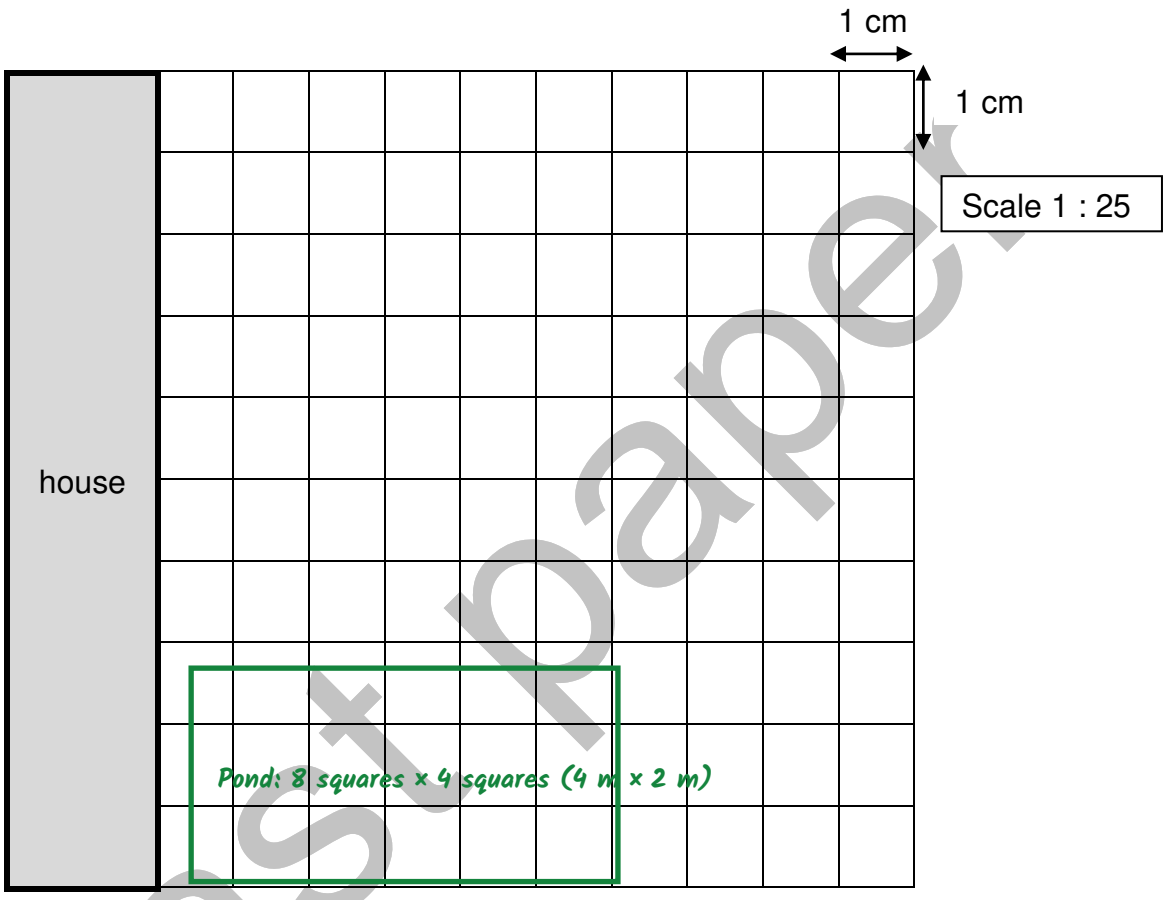
© Training Qualifications UK 2023 All Rights Reserved

6. Charlie wants to build a pond in a garden.

The pond needs to be a rectangle with:

- a length of 2 metres and a width of 1 metre
- **at least** 1 metre away from the house.

Draw a possible position for the pond on the scale diagram below:



[2]

Past paper

Scale: 1 square = 0.5 m --> 4 m = 8 squares, 2 m = 4 squares

Positioned at least 2 m (4 squares) away from the house [2 marks]

 © Training Qualifications UK 2023 All Rights Reserved

7. Last year 825 500 people visited a museum.

The manager is set a target to increase the number of visitors by 22%.

The manager thinks this will be more than 1 005 000 visitors.

Is the manager correct?

Show how you decide.

Step 1: Increase by 22% = × 1.22

825 500 × 1.22 = 1 007 110

Comparison: 1 007 110 > 1 005 000. Therefore Yes.

Answer Yes (1 007 110 > 1 005 000) [3 marks]

[3]

Past paper

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

8. Leslie needs to put a fence around a pond.

The pond is in the shape of a circle with a diameter of 4.26 meters.

Leslie buys 13.5 metres of fencing.

Has Leslie bought enough fencing?

Show how you decide.

Use π = 3.14

Formula: Circumference = π × d

Circumference = 3.14 × 4.26 m = 13.3764 m

Comparison: 13.3764 m < 13.5 m. Therefore Yes, Leslie has enough.

Answer Yes (13.38 m < 13.5 m) [3 marks]

[3]

Past paper

Master Functional Skills maths - TQUK Mathematics Level 2 Past Paper 3 Solutions © Training Qualifications UK 2023 All Rights Reserved

9. Taylor books a holiday.

The next day, their friend books the same holiday for £690 **after** receiving a 4% discount.

How much more did Taylor pay for the holiday?

Step 1: £690 represents (100% - 4%) = 96% of original price

Original price = £690 ÷ 0.96 = £718.75

Money saved = £718.75 - £690 = £28.75

Answer £ £28.75 ✓ [3 marks]

[3]

Past paper

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

10. Chen sells cakes.

The table below shows the ingredients in one cake:

Ingredient	Grams
Flour	275
Sugar	200
Baking powder	5

A customer wants to know what fraction of the cake is sugar.

Chen tells the customer $\frac{7}{12}$

Is this correct?

Show how you decide.

Total ingredients = 275 g + 200 g + 5 g = 480 g

Fraction of butter = 200 ÷ 480 = 5/12

Comparison: 5/12 ≠ 7/12 (5/12 is less than 7/12). Therefore No.

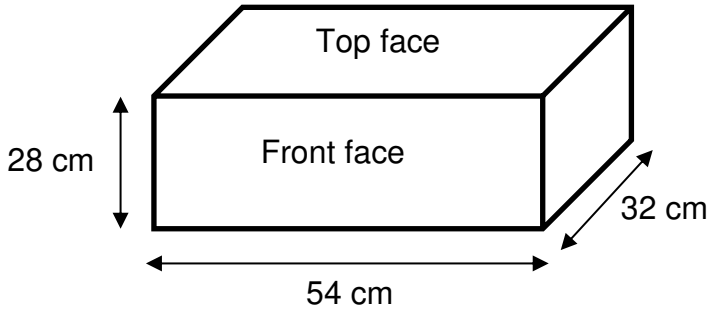
Answer No (fraction is 5/ ✓ [3 marks]

[3]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

11. Lee wants to paint the front face and top face of this cuboid as part of an art project:



Not drawn to scale

Lee will paint yellow, green and white stripes in the ratio 5 : 3 : 2

How much of the cuboid will be painted yellow?

Give your answer in square centimetres (cm^2).

Base = 54 × 28 = 1512 cm^2 / 2 sides = 2 × (32 × 28) = 1792 cm^2

Total Surface Area = 3240 cm^2

Half-area = 3240 ÷ 10 × 5 = 1620 cm^2

Answer 1620 c ✓ [4 marks] **cm^2**

[4]

Questions continue on the following page

Past paper

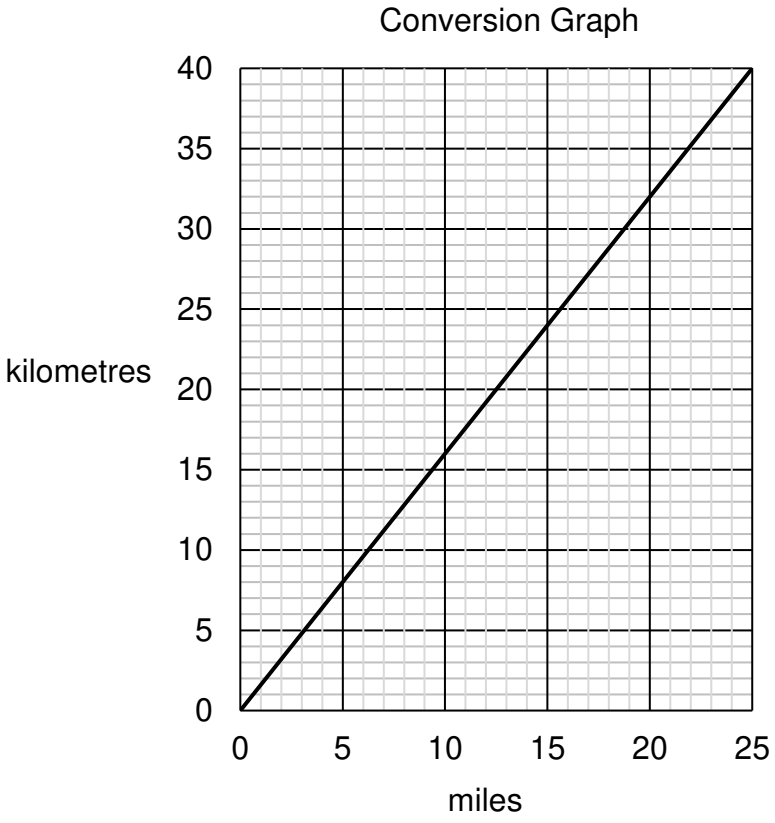
 © Training Qualifications UK 2023 All Rights Reserved

12. Harper is planning a road trip with this information:

- the total distance is 1600 **kilometres**
- Harper only wants to drive a maximum of 225 **miles** each day.

How many whole days will Harper need to complete the 1600 km?

Use this graph to convert between kilometres and miles.



Show your working.

Conversion from graph: 8 km = 5 miles (or 40 km = 25 miles)

1600 km = (1600 ÷ 8) × 5 = 1000 miles

Days = 1000 ÷ 225 = 4.444... days --> Rounds up to 5 days

Answer 5 days [4 marks] **days**

[4]

© Training Qualifications UK 2023 All Rights Reserved

13. Over a period of six weeks, Jamie sold chocolate cupcakes and lemon cupcakes.

The number of chocolate cupcakes Jamie sold each week is shown below:

Number of chocolate cupcakes sold					
Week 1	**Week 2**	**Week 3**	**Week 4**	**Week 5**	**Week 6**
91	85	64	68	100	72

The number of lemon cupcakes Jamie sold is summarised below:

Lemon cupcakes sold	
Mean number sold per week	84
Range	20

Jamie claims:

a) "On average, I have sold more chocolate cupcakes per week"

b) "The chocolate cupcake sales are more consistent".

Are both of Jamie's claims correct?

Give reasons for your answers.

Show working here:
Reason - claim a):
Reason - claim b):

[4]

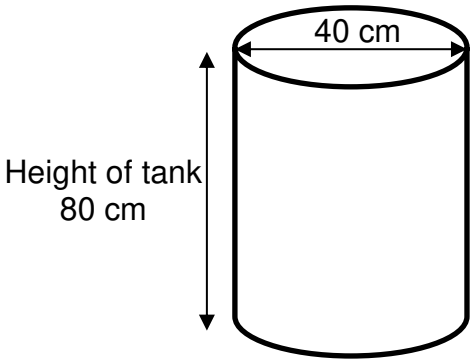
Questions continue on the following page

Past paper

© Training Qualifications UK 2023 All Rights Reserved

14. Ali is setting up a saltwater fish tank.

The fish tank is in the shape of a cylinder.



Not drawn to scale

Ali uses this formula to work out how many grams of salt to add to the tank:

$$\text{Amount of salt in grams} = \left(\frac{\text{volume of tank}}{1000}\right) \text{multiplied by } 35$$

Ali thinks the tank will need more than 3.75 kilograms of salt.

Is Ali correct?

Show how you decide.

Use π = 3.14

Mean for Lemon: (91+85+68+104+84+48) ÷ 6 = 480 ÷ 6 = 80

Range for Lemon: 104 - 48 = 56 (Orange range is 70 - 45 = 25)

Comment: No, Lemon had a higher mean (80 > 56) and higher variation.

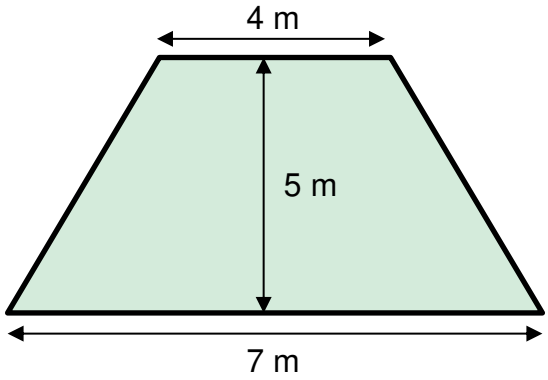
Answer No (Mean = 80, Ran... S... [4 marks]

[5]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

15. Riley wants to estimate how much it will cost to lay a patio in the garden below:



Not drawn to scale

Riley finds these prices online:

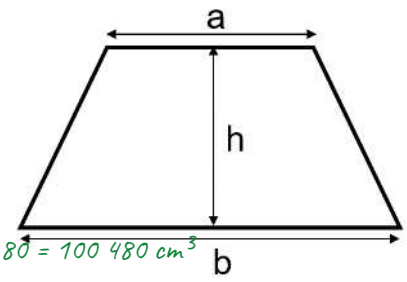
Price per square metre (m^2)					
£85	£100	£75	£130	£150	£135

Riley uses the median price to estimate how much it will cost to lay the patio.

How much does Riley estimate the patio will cost?

Use:

$$\text{Area} = \frac{(a+b)h}{2}$$



Cylinder Volume = $\pi \times r^2 \times h = 3.14 \times 20^2 \times 80 = 100\,480\ cm^3$

Mass = (100 480 ÷ 1000) x 35 g = 3516.8 g = 3.5168 kg

Comparison: 3.5168 kg < 3.75 kg. Therefore No.

Show your working.

Answer box is on the next page

Past Paper

© Training Qualifications UK 2023 All Rights Reserved

Q14 Conclusion: 3.517 kg < 3.75 kg --> No

No (3.52 < 3.75 kg) [2 marks]

Trapezium Area = ((4 + 7) ÷ 2) × 5 = 5.5 × 5 = 27.5 m²

Ordered costs: 75, 85, 100, 130, 135, 150 --> Median = (100 + 130) ÷ 2 = £115

Total Cost = 27.5 m² × £115 = £3162.50

Answer	£3162. [6 marks]

[6]

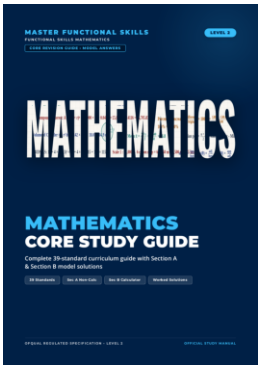
Past paper

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Maths Level 2 Exam



Core Curriculum Manual

Maths Level 2
Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

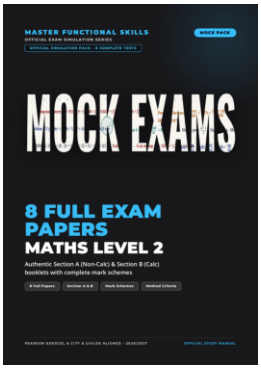


1,000+ Practice Drills

Maths Level 2
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF



Realistic Mock Papers

Maths Level 2
Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF



Pocket Flashcards A6

Maths Level 2
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by NHS trusts, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk